20/7/26 EXPERIMENT - 1

BASIC NETWORKING COMMANDS

AIM: Study of various network commands used in linux and windows

i) arp -a :

ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router

Output:

i) (172.16.9.212) at 7C:57:58:38:26:24 [ether] on enp 350

ii) (172.16.8.249) at 00.27.02:13:15:77 [ether] on enp 350

(iii) (172.10.11.207) at 7C:57:58:38:36:87 [ether] on enp 350

(iv) (172.16.8.131) at 00:21:0e:13:f4:41 [ether] on enp 350

(v) (172.16.8.145) at 00.27.0e.13:ed:44 [ether] on enp 350.

2) hostname

This is the simplest of all TCP/IP commands of simply displays the name of your computers.

output:

localhost.localdomain

3) netstat:

netstat displays a variety of statics about a computers active TCP/IP connection it is command line tool for monitoring network connection both incoming and outgoing as well as viewing routing tables interface and non increment.

Output:

Proto	Recv-Q	Send-Q	LA	FA	State
tcp	0	0	localhost localdrama.ash	172.16.8.135.56	Establ
tcp	0	0	localhost localdrama.ash	172.16.8.135.5a1	Estabh
tcp	0	0	localhost localdrama.ash	172.16.8.11:5ac29	Estabh

4) nslookup:

It is a tool used to perform DNS lookups in linux, it is used to display DNS details such as IP address of a particular computer, The MX records

Eg: nslookup www.google.com.

Output:

Server : 172.16.8.1

address : 172.16.8.1 #53

www.snapchat.com

Canonical name web.gcp.api
snapchat.com

web-gcp.api.snapchat.com

Name : web-gcp.api sc.qc.com

address: 34.149.46.130

5) ping (packet internet groper)

Command is the best way to test connection between 2 nodes, ping use to communicate to each other devices

i) # ping hostname (ping: localhost)

output:

64 bytes from localhost (::1):
icmp_seq = 1 #L = 64
time = 0.77 ms

64 bytes from localhost (::1):
icmp_seq 2 #L = 64
time = 0.36 ms

64 bytes from localhost (::1):
icmp_seq = 3 HC = 64
time = 0.035 ms

64 bytes from localhost (::1)
icmp_seq = 4 HC = 64
time = 0.036 ms

6. Route:

Route command is used to show manipulate the IP raising table. It is primarily setup to static route to specify host.

Output: Destination	Gateway	Genmask	Flag	Metrics
default	gateway	0.0.00	0-67	100
172.16.8.0	0.0.0.0	255.255.2520	U	100

7. IP

The IP command is one of the basis command every administrator will need in daily work, from setting up new system and assigning IP to trouble the IP command can show address information, manipulate routing, plus display network various device and tunnels.

Output:

Click /loopback 00:00 :00:00:00 brd 00:00:00:00:00:00

Click 127.0.0.1 /8 scope host lo valid lft forever preferred lft lower

8) Ipconfig:

The Ipconfig command was a staple in many sysadmins bet for configuration and trouble shooting network it has discussed above.

Output:

wipsso. flag 4099 <UP, BROADCAST, MULTICAST>
MNL500

ether 16:sd :7f:bd :46:86 ; hequrlan 1000

ex packets obytes o (oob)

ex error o dropped o overruns o name o

TX packets o bytes o (0.0B)

TX errors o dropped overome o
collisions o

9) Mtr

MTR is a program with a command line interface that server as network diagonistic and double shing tool.

Output:

-F, --filename FILE read hostname (S) form

-U, _udp use UDP instead of icmp echo

-T, --tcp use TCP instead of ICMP echo

-I, -- para port target port no for tcp, bchp

-i, -- interval SECONDS ICMP echo request interval.

9) tcpdump:

The tcpdump command is designed for capturing and displaying packets

Output:

1) eppzso (-p, running, connected)
2) any (pseudo device captures on all interface)
3) lo [up, running, loopback]
4) wlp350 [up, wireless, not associated]
5) usbmon [raw USB traffic, bus number]

10) ping

ping is a tool that verifies IP level connectivity to another TCP/IP by sending internet control message protocol echo request message

ip:

64 bytes from 130.46.149.34 to google user content.com

64 bytes from 130.46.146.34 bc google user content

com (34.149.46.130) icmp :seq =1

=reno time =137 ms